BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (12/10/2026 – 18/10/2026)

WEEK 12

PROJECT NAME	CHATBOT FOR STRUCTURED, UNSTRUCTURED AND SEMANTIC KNOWLEDGE QUERY

PROJECT DESCRIPTION :	DESIGN AND IMPLEMENT A CHATBOT THAT CAN ANSWER TO DIFFERENT TYPE OF QUERY LIKE STRUCTURED, UNSTRUCTURED AND SEMANTIC KNOWLEDGE BY DECIDING OWN WHAT TYPE OF QUERY BY
GROUP MEMBER :	PRATYUSH ANAND, ADITYA SINGH
GROUP ID :	2026G306
GROUP GUIDE :	VEDANT JOSHI
GROUP COORDINATOR :	SIDHDHARTH PATEL
COLLEGE GUIDE NAME :	EHT SHAM
COLLEGE GUIDE No. :	+91 9818938592
COLLEGE GUIDE EMAIL. :	Ehte.sham@pilani.bits-pilani.ac.in
COLLEGE NAME :	BITS PILANI

12/10/2026	• Reviewed the chatbot's end-to-end workflow and refined the unified organization knowledge pool for documents, images, and structured data. • Planned local testing with multiple organizations, focusing on evidence-based answers, source selection, and separate conversation histories.
13/10/2026	• Tested image-based questions using an AQI screenshot and identified incomplete extraction, missing categories, and incorrect support labels. • Investigated organization-specific document access and conversation behavior to identify possible cross-organization leakage.
14/10/2026	• Configured and troubleshot the local runtime using Ollama, Qwen, embeddings, and a neural reranker. • Improved AQI table extraction and answer validation; tested category boundaries, complete summaries, and refusal of unsupported current readings.
15/10/2026	• Refined requirements for answering natural-language questions across the organization's knowledge pool without requiring filenames. • Defined file-selection behavior and local diagnostics requirements for tracking extraction, retrieved evidence, model prompts, responses, and validation decisions.
16/10/2026	• Tested real PDF and CSV uploads, including the INSPIRE_MANAK dataset containing 43,381 records; independently checked counts, totals, and filtered results. • Used developer traces to identify incorrect rejection of valid SQL answers and specified improvements to upload review, dataset context, and overlapping chat-interface elements.
17/10/2026	• Verified fixes for SQL evidence validation and complete-record lookups, including readable answers, citations, missing identifiers, and records beyond the indexed sample. • Tested grouped ranking questions and diagnosed a validator incorrectly rejecting correct SQL and replacing it with an overall **count**; prepared requirements for correcting source resolution and ranking validation.
18/10/2026	**Holiday (Sunday)**

Next Week Plan	Improve query validation and UI, and test accuracy across datasets and organizations.

REFERENCE:

https://microsoft.github.io/graphrag/get_started/

https://microsoft.github.io/graphrag/index/default_dataflow/

https://docs.ollama.com/

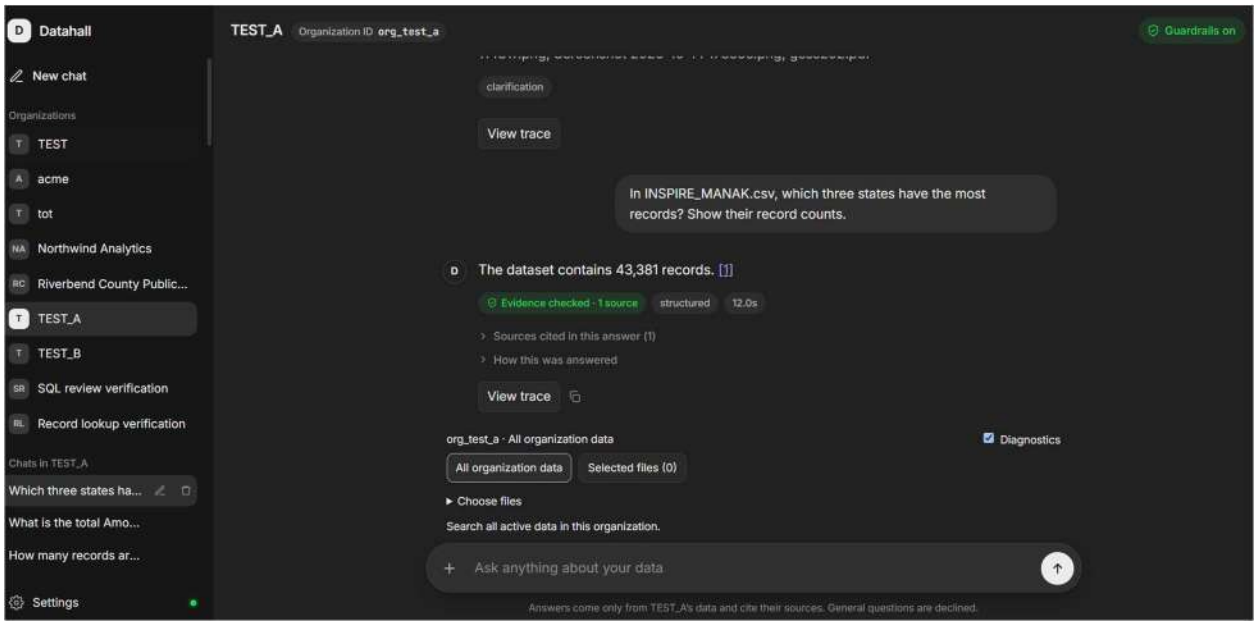
https://www.sqlite.org/docs.html

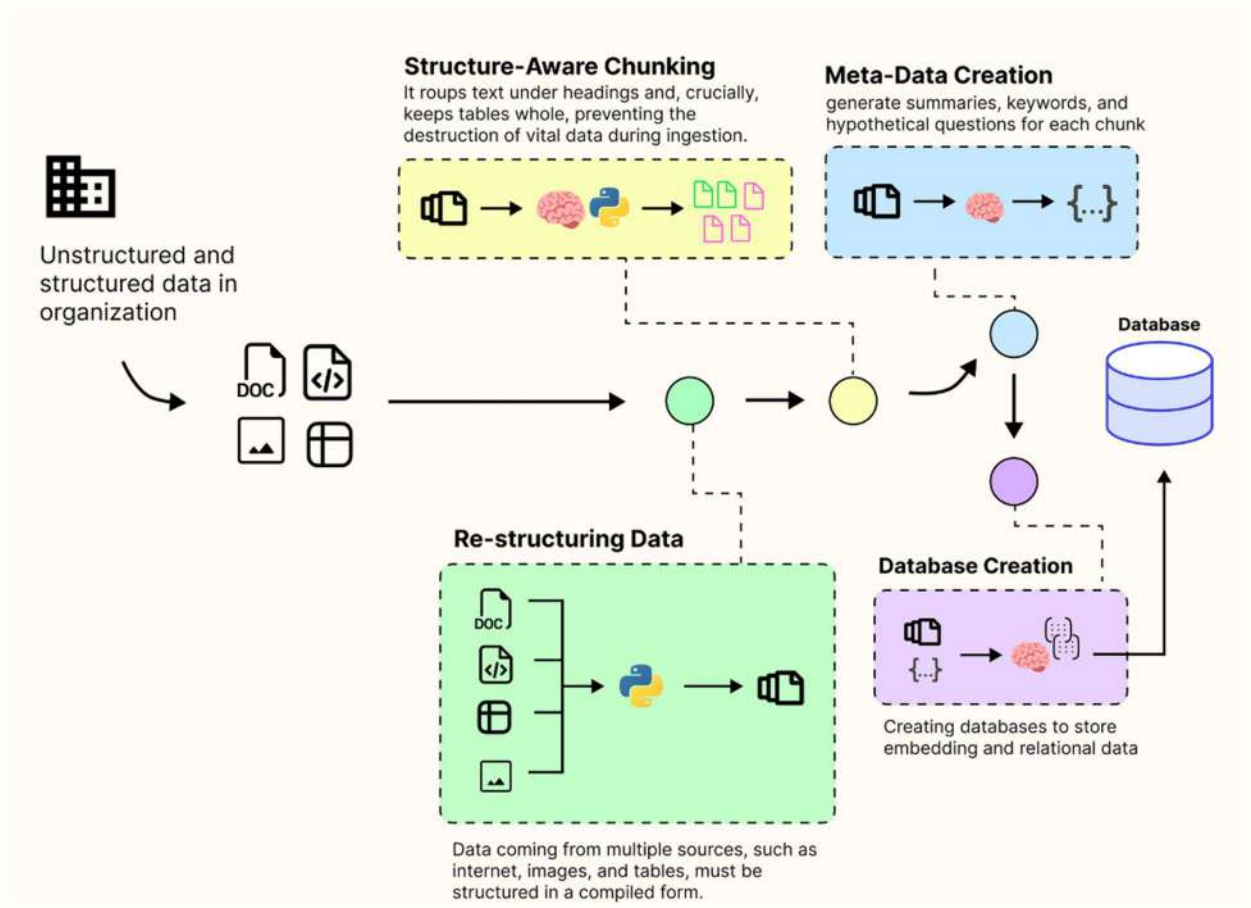
https://github.com/meta-llama/llama3

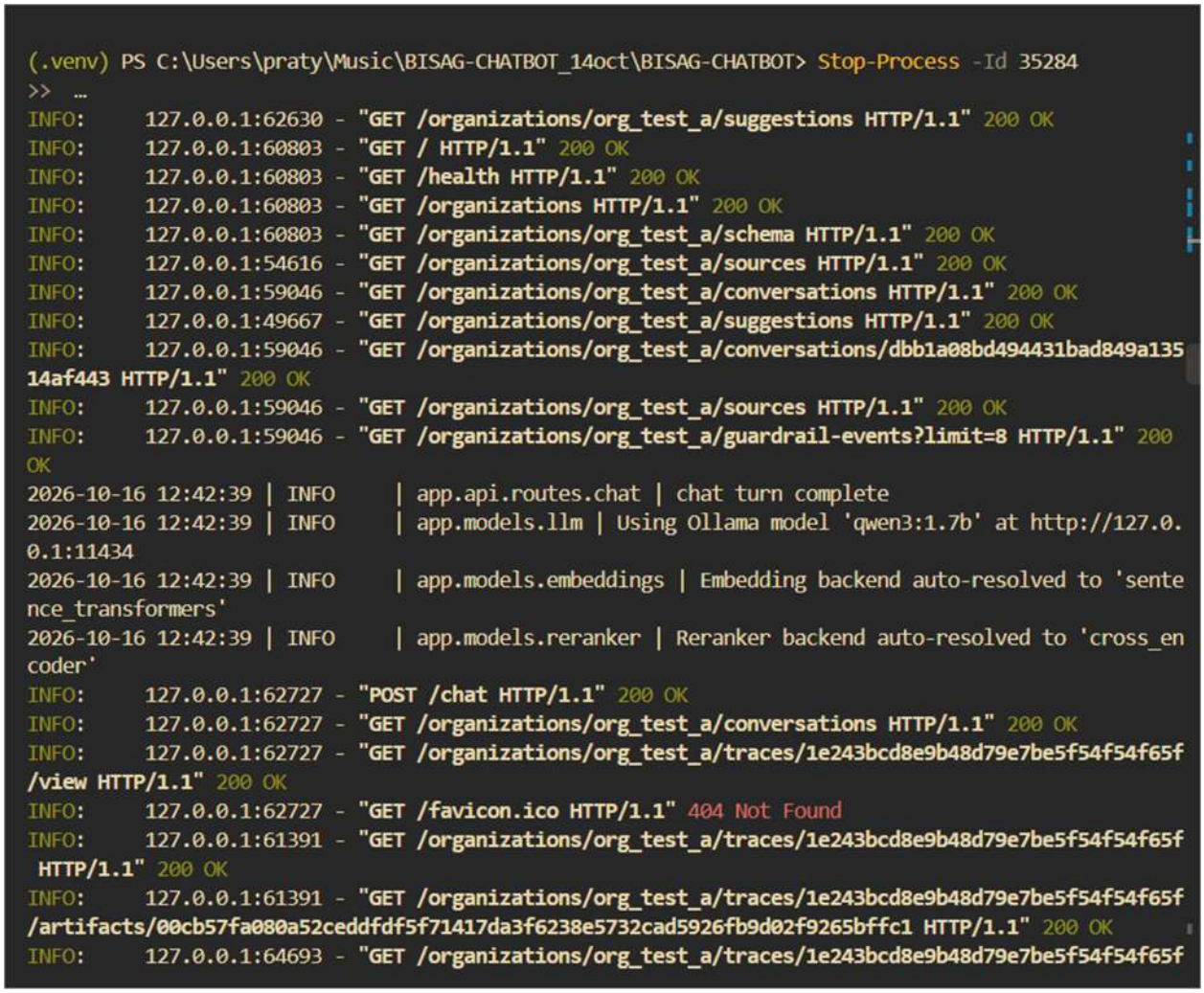
https://docs.python.org/3/library/sqlite3.html

https://github.com/QwenLM/Qwen2.5-VL

SCREEN SHOTS:




Structure-Aware Chunking
It roups text under headings and, crucially, keeps tables whole, preventing the destruction of vital data during ingestion.
Meta-Data Creation
generate summaries, keywords, and hypothetical questions for each chunk
Unstructured and structured data in organization
DOC
Database
Re-structuring Data
DOC
Database Creation
Creating databases to store embedding and relational data
Data coming from multiple sources, such as internet, images, and tables, must be structured in a compiled form.



```
(.venv) PS C:\Users\praty\Music\BISAG-CHATBOT_14oct\BISAG-CHATBOT> Stop-Process -Id 35284
>>  …
INFO:     127.0.0.1:62630 - "GET /organizations/org_test_a/suggestions HTTP/1.1" 200 OK
INFO:     127.0.0.1:60803 - "GET / HTTP/1.1" 200 OK
INFO:     127.0.0.1:60803 - "GET /health HTTP/1.1" 200 OK
INFO:     127.0.0.1:60803 - "GET /organizations HTTP/1.1" 200 OK
INFO:     127.0.0.1:60803 - "GET /organizations/org_test_a/schema HTTP/1.1" 200 OK
INFO:     127.0.0.1:54616 - "GET /organizations/org_test_a/sources HTTP/1.1" 200 OK
INFO:     127.0.0.1:59046 - "GET /organizations/org_test_a/conversations HTTP/1.1" 200 OK
INFO:     127.0.0.1:49667 - "GET /organizations/org_test_a/suggestions HTTP/1.1" 200 OK
INFO:     127.0.0.1:59046 - "GET /organizations/org_test_a/conversations/dbb1a08bd494431bad849a135
14af443 HTTP/1.1" 200 OK
INFO:     127.0.0.1:59046 - "GET /organizations/org_test_a/sources HTTP/1.1" 200 OK
INFO:     127.0.0.1:59046 - "GET /organizations/org_test_a/guardrail-events?limit=8 HTTP/1.1" 200
OK
2026-10-16 12:42:39 | INFO     | app.api.routes.chat | chat turn complete
2026-10-16 12:42:39 | INFO     | app.models.llm | Using Ollama model 'qwen3:1.7b' at http://127.0.
0.1:11434
2026-10-16 12:42:39 | INFO     | app.models.embeddings | Embedding backend auto-resolved to 'sente
nce_transformers'
2026-10-16 12:42:39 | INFO     | app.models.reranker | Reranker backend auto-resolved to 'cross_en
coder'
INFO:     127.0.0.1:62727 - "POST /chat HTTP/1.1" 200 OK
INFO:     127.0.0.1:62727 - "GET /organizations/org_test_a/conversations HTTP/1.1" 200 OK
INFO:     127.0.0.1:62727 - "GET /organizations/org_test_a/traces/1e243bcd8e9b48d79e7be5f54f54f65f
/view HTTP/1.1" 200 OK
INFO:     127.0.0.1:62727 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:61391 - "GET /organizations/org_test_a/traces/1e243bcd8e9b48d79e7be5f54f54f65f
 HTTP/1.1" 200 OK
INFO:     127.0.0.1:61391 - "GET /organizations/org_test_a/traces/1e243bcd8e9b48d79e7be5f54f54f65f
/artifacts/00cb57fa080a52ceddfdf5f71417da3f6238e5732cad5926fb9d02f9265bffc1 HTTP/1.1" 200 OK
INFO:     127.0.0.1:64693 - "GET /organizations/org_test_a/traces/1e243bcd8e9b48d79e7be5f54f54f65f
```